

19 More Bayesian Analysis of a Numerical Variable

Example 19.1. How late do people arrive at parties? [FiveThirtyEight](#) conducted a survey to address this question. We'll assume this is a reasonably representative sample and arrival times are measured reliably. Arrival times are measured in minutes after the scheduled start time, rounded to the nearest minute. (Negative arrival times represent arrivals before the scheduled start time.)

1. What would you expect the population distribution of party arrival times to look like? For example, what percent of arrivals would you expect to be early (negative)? Later than two hours? What would you expect of other features like shape, center, and variability?
2. We will start by assuming that, given the values of relevant parameters, arrival times follow a conditional Normal distribution. Describe what this assumption says about arrival times. What are the relevant parameters, and what do they represent? (We will revisit this conditional Normal assumption later, but go with it for now.)
3. Suggest a prior distribution. Explain your reasoning.
4. Describe how you could simulate the prior predictive distribution.

5. This [applet](#) conducts the simulation from the previous part. Move the sliders to enter your prior mean and SD for μ and σ . Does the distribution of arrival times seem reasonable based on your expectations? For example, what percent of arrivals are early (negative), or later than two hours — and do these values seem reasonable? If not, revise your assumptions and try again (i.e., play with the sliders in the applet) until you find a distribution that is reasonable. Do not worry about getting it perfect; you just want to settle on assumptions that provide a reasonable starting point. (The assumptions include both the conditional Normal model for arrival times and the prior distribution of parameters. For now we are focusing on the prior distribution, so take the conditional Normal model as given. We'll revisit that assumption later.)

Example 19.2. Section [19.1.2](#) summarizes the FiveThirtyEight sample¹ of 803 arrival times (rounded to the nearest minute).

We should always first explore the sample data, but to illustrate a point we will not do that yet. Instead, we'll skip straight to fitting the model.

1. Use `brms` to fit the model and summarize the posterior distribution. Briefly describe the posterior distribution.
2. Simulate the posterior predictive distribution. Briefly describe the posterior predictive distribution.

¹We couldn't find the raw FiveThirtyEight data online, so this data was constructed to match the histogram in the article.

3. Perform a visual posterior predictive check. What does the posterior predictive check say about the appropriateness of our model? Could any problems be fixed by switching from a Normal to a t likelihood?

Example 19.3. The sample data suggests a model that allows for skewness would be more appropriate than Normal. This primarily results in a change of the likelihood function. There are several likelihood functions we can use, but we'll try a [Skew Normal](#) family, which introduces an additional skewness parameter α . Since α is a parameter it also needs a prior distribution. Rather than redoing our prior predictive tuning, we'll rely on the fact that software packages like `brms` can utilize default priors.

1. Use `brms` to fit the skew Normal model, letting `brms` choose the prior for α (but specifying the priors for μ and σ as before). What prior does `brms` use?
2. Summarize and describe the posterior distribution.
3. Use simulation to approximate the posterior predictive distribution. Summarize and describe the posterior predictive distribution.
4. Perform a visual posterior predictive check. What does the posterior predictive check say about the appropriateness of the model? Which model seems to be a better fit for the data: likelihood based on Normal or Skew-Normal?

5. Fit the model to the data in `brms` using the `skew_normal` family but letting `brms` choose the prior for all parameters. What prior does `brms` choose? Is the posterior sensitive to the choice of prior?
6. Are the posterior distributions of parameters skewed or relatively symmetric? Is the posterior predictive distribution skewed or relatively symmetric?
7. Write a few sentences summarizing the conclusions of the Bayesian model in context.

19.1 Notes

19.1.1 Prior predictive distribution (Normal likelihood)

```
n_rep = 10000  
  
mu_sim = rnorm(n_rep, 40, sd = 7)  
  
sigma_sim = pmax(0.001, rnorm(n_rep, 20, 5))  
  
y_sim = rnorm(n_rep, mu_sim, sigma_sim)  
  
ggplot(data.frame(y_sim),  
       aes(x = y_sim)) +
```